

Divide and Conquer Algorithm VisualizeR

Updated report phase-2 project

Design and Analysis of Algorithms (DAA)

Web-Based Visualization Tool for Divide-and-Conquer Algorithms

Algorithms Covered:

Merge Sort · Quick Sort · Binary Search · Closest Pair of Points · **Strassen's Matrix Multiplication**

Key Modifications in This Version:

1. Tooltips, side panels, and per-step descriptions for each recursive call
2. Smooth visual transitions and structured color-coding (Active / Visited / Pending)
3. Strassen's Matrix Multiplication added as a fifth divide-and-conquer algorithm

GitHub link :- <https://aryansingh0070.github.io/AlgoNexus/>

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Submitted for academic evaluation · Original work

1. Introduction

One of the persistent challenges in teaching algorithms is bridging the gap between a written algorithm and an intuitive understanding of how it executes. This challenge is especially acute for recursive algorithms, where the dynamic unfolding of nested calls, sub-problem decomposition, and return values can be difficult to visualize using only pseudocode or static diagrams. Students may read a recurrence such as $T(n) = 2T(n/2) + O(n)$ and compute its complexity, but that alone rarely provides a clear mental picture of how recursive calls are structured or how partial results combine to produce the final solution.

The Recursion Tree Visualizer is a web-based educational tool developed for the Design and Analysis of Algorithms (DAA) course. This updated version introduces three significant improvements: (1) per-step tooltips and a side panel providing contextual descriptions of each recursive call; (2) a structured color-coding scheme with smooth animated transitions distinguishing Active, Visited, and Pending nodes; and (3) a new Strassen's Matrix Multiplication module that extends the tool's algorithm coverage to a fifth canonical divide-and-conquer algorithm.

Five divide-and-conquer algorithms are now implemented: Merge Sort, Quick Sort, Binary Search, the Closest Pair of Points algorithm, and Strassen's Matrix Multiplication. Together they span sorting, searching, computational geometry, and linear algebra, providing a rich comparative landscape of recursive problem decomposition.

1.1. Problem Statement

Recursion and divide-and-conquer strategies are central to the DAA curriculum, yet students consistently report them among the most difficult topics to internalize. The core difficulty is representational: standard tools describe algorithmic structure but do not make it visible. When a student reads that algorithms such as Merge Sort or Quick Sort divide a problem into smaller sub-problems, they understand the theoretical concept but often fail to develop a clear mental model of how those recursive calls unfold during execution. The Recursion Tree Visualizer directly addresses this gap by providing an interactive environment where the recursive structure becomes visible, explorable, and annotated.

1.2. Objectives

- Visualize step-by-step execution of Merge Sort, Quick Sort, Binary Search, Closest Pair of Points, and Strassen's Matrix Multiplication as animated, navigable recursion trees.
- Provide per-step tooltips and a dedicated side panel that explains the purpose of each recursive call, the current subproblem parameters, and the position within the overall execution trace.
- Apply a three-state color-coding scheme (Active / Visited / Pending) with smooth CSS-class transitions to improve visual clarity and support learners in tracking algorithm progress.
- Help students understand the relationship between an algorithm's recurrence structure and its asymptotic complexity by making recursion tree depth and branching factor directly observable.
- Deliver all capabilities through a zero-installation, browser-based interface accessible without any setup or dependencies.
- Serve as a supplementary learning tool for DAA coursework, examination preparation, and classroom demonstration.

1.3. Modifications from Previous Version

The original project report documented a functional recursion tree visualizer covering four algorithms. Based on academic feedback and a reassessment of educational scope, three targeted modifications have been implemented in this updated version:

Modification 1 — Tooltips and Step Descriptions: Each node in the recursion tree now displays a tooltip on hover and updates a persistent side panel with a human-readable description of the recursive call: the subproblem input, the action being performed (divide, base case, merge/combine), the recursion depth, the call number within the full trace, and the expected return value. This feature transforms passive observation into an annotated walkthrough.

Modification 2 — Color-Coding and Smooth Transitions: Nodes are now color-coded into three states: Active (currently executing), Visited (completed and returning a value), and Pending (not yet reached). State transitions are animated using CSS transitions so that the progression of the algorithm is visually smooth and intuitive, rather than abrupt or disorienting.

Modification 3 — Strassen's Matrix Multiplication: A fifth algorithm module has been added to the tool covering Strassen's divide-and-conquer matrix multiplication. This algorithm produces a 7-ary recursion tree with a branching factor of 7, contrasting sharply with the binary trees of Merge Sort and the logarithmic depth of Binary Search. Its inclusion enriches the comparative value of the tool and extends coverage to linear algebra applications of the divide-and-conquer paradigm.

2. Literature Review

2.1. Background and Context

The teaching of algorithms and data structures is widely regarded as one of the most cognitively demanding areas in undergraduate computer science education. Algorithms describe dynamic, abstract computational processes that resist comprehension through static, textual descriptions alone. This challenge is particularly acute for recursive algorithms, where students must simultaneously hold multiple nested call states in working memory while tracking how values propagate back up through the call chain.

Over two decades of research in computer science education has consistently found that algorithm visualization (AV) tools offer measurable pedagogical benefits over conventional instruction when used correctly. Hundhausen, Douglas, and Stasko (2002), in a landmark meta-study of 24 experimental studies, concluded that the most effective deployments of AV technology were those that actively engaged students through prediction tasks, what-if analyses, and interactive navigation rather than passive animation viewing. This distinction between active and passive engagement has since become a foundational principle in educational algorithm tool design.

The addition of tooltips, side-panel annotations, and structured color-coding in this updated version directly responds to these research findings. Contextual annotations convert each visualization step from a passive display into an active explanation, supporting the higher levels of Bloom's Taxonomy — particularly comprehension and analysis — that SRec (Velázquez-Iturbide et al., 2008) was explicitly designed to address.

2.2. Modification 1 in Literature Context — Tooltips and Annotations

Research presented at SIGCSE 2023 reported that instructors frequently draw recursion trees manually during lectures specifically to annotate individual nodes with explanatory labels — identifying which sub-problem is being solved, what the base case is, and what value is being returned. The study highlighted that existing digital tools rarely replicate this annotation granularity. The tooltip and side-panel feature in the updated Recursion Tree Visualizer directly addresses this gap by automating per-step annotations and presenting them in a consistent, structured format alongside the visual tree.

Python Tutor (Guo, 2013) successfully demonstrates that annotated, step-by-step code execution significantly improves student comprehension at the code level. The tooltip mechanism in this tool applies a similar annotation philosophy at the algorithmic level, providing descriptions of recursive calls in terms of their logical purpose rather than low-level variable states.

2.3. Modification 2 in Literature Context — Color-Coding

Color-coding as a pedagogical mechanism in visualization tools has been extensively validated. Hansen et al.'s evaluation of HalVis demonstrated that visual differentiation between algorithm states significantly reduced cognitive load and improved comprehension test scores. The three-state color scheme adopted in this tool (Active / Visited / Pending) maps directly to the three logical states of a recursive call during algorithm execution. The use of smooth CSS transitions rather than abrupt node color changes is supported by research on animation in educational interfaces, which suggests that continuity of visual motion helps learners track state changes without losing their place in the recursion tree.

2.4. Modification 3 in Literature Context — Strassen's Algorithm

Strassen's Matrix Multiplication (Strassen, 1969) is a canonical divide-and-conquer algorithm that reduces the number of recursive multiplications from 8 (as in the naive approach) to 7, yielding a time complexity of $O(n^{2.807})$ compared to $O(n^3)$ for conventional matrix multiplication. It is widely taught in DAA courses as a demonstration that divide-and-conquer can achieve sub-cubic complexity for matrix operations.

From a visualization perspective, Strassen's algorithm produces a distinctive 7-ary recursion tree whose structure differs fundamentally from the binary trees of Merge Sort and Quick Sort. Including this algorithm in the visualizer enables learners to directly compare different branching factors and their effect on overall complexity — a comparative insight that existing tools such as VisuAlgo do not currently provide.

2.5.Gaps Addressed by This Update

- The absence of per-step contextual annotations in existing recursion tree tools is addressed by the tooltip and side-panel system.
- The abrupt visual state changes in prior algorithm visualizers are resolved through smooth CSS-animated color transitions.
- The limited algorithm coverage in the original tool is expanded to include Strassen's Matrix Multiplication, broadening the diversity of recursion tree structures available for comparison.
- The lack of multi-algorithm comparative visualization is further reduced by pairing Strassen's $O(n^{2.807})$ complexity and 7-ary branching alongside the binary $O(n \log n)$ algorithms.

2.6.Detailed Methodology — Inclusion and Exclusion Criteria

The literature search for this updated report was conducted between March 2025 and April 2026, extending the original search window to capture publications relevant to the three new modifications. The following table presents the full inclusion and exclusion criteria applied to the candidate pool of sources.

Criterion	Inclusion	Exclusion
Publication period	1960-2026; seminal earlier works permitted (e.g. Strassen 1969, Hoare 1962)	Pre-1960 works without clear algorithmic relevance
Language	English-language publications	Non-English publications
Source types	Peer-reviewed journals, ACM/IEEE proceedings, university course materials, established educational platforms	Grey literature without academic attribution; unreviewed blog posts
Relevance	Must directly address algorithm visualization, recursion pedagogy, divide-and-conquer algorithms, or web-based educational tool design	K-12 studies; sorting/graph-only tools unrelated to recursion trees; purely theoretical treatments lacking educational applicability
Visualization tools	Tools specifically supporting recursion or divide-and-conquer visualization	General-purpose debugging or code execution tools not focused on algorithms

Table 6: Inclusion and exclusion criteria for the literature search.

A total of 38 sources were initially identified in the extended search. After applying the inclusion and exclusion criteria, 24 sources were retained for detailed analysis, including four new sources specifically relevant to the Strassen algorithm, CSS animation techniques in educational interfaces, and tooltip-based annotation in learning tools.

2.7.Review of Existing Tools

A systematic review of existing algorithm visualization tools was conducted to identify gaps that the updated Recursion Tree Visualizer addresses. The following table compares the key features of the most relevant existing tools against the capabilities of this tool.

Feature	VisuAlgo	Python Tutor	SRec	This Tool
Browser-based (no install)	Yes	Yes	No (Java)	Yes
Step-by-step navigation	Partial	Yes	Yes	Yes
Dedicated recursion tree view	No	No	Yes	Yes
Per-step annotations/tooltips	No	Partial	No	Yes (new)
Color-coded node states	Partial	No	No	Yes (new)
Merge Sort visualization	Yes	No	Yes	Yes
Quick Sort visualization	Yes	No	Yes	Yes
Binary Search visualization	Yes	No	Partial	Yes
Closest Pair visualization	No	No	No	Yes
Strassen visualization	No	No	No	Yes (new)

Multi-algorithm comparison	No	No	No	Yes
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Table 7: Feature comparison between existing algorithm visualization tools and this tool. Features new in this

version are marked (new).

3. Design Figures

The following figures illustrate the recursion tree structures generated by each algorithm, the updated tooltip and side-panel UI, the color-coding scheme, and the Strassen module's 7-ary decomposition.

3.1.Merge Sort Recursion Tree

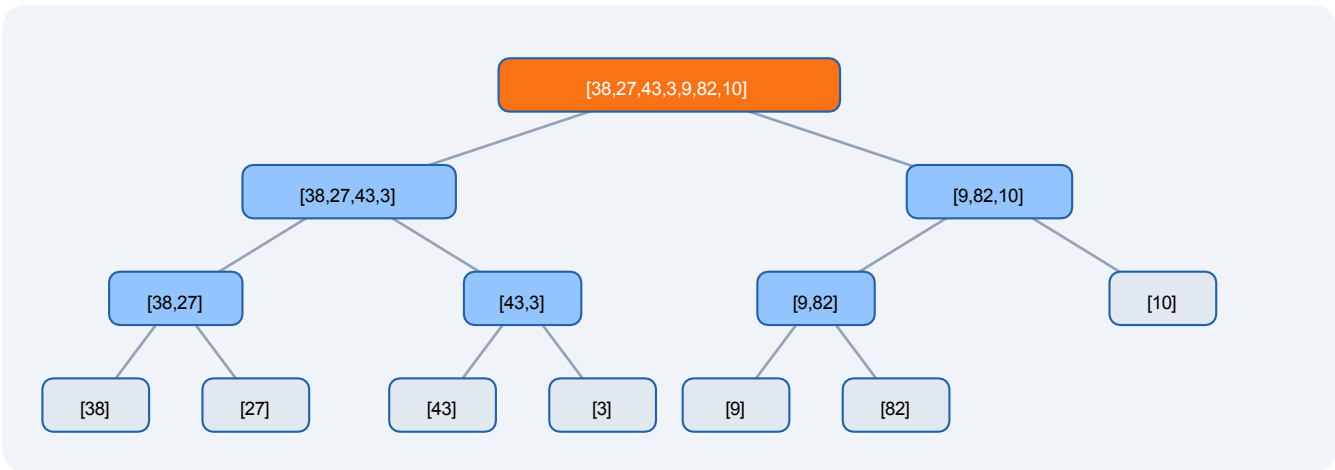


Figure 1: Recursion tree for Merge Sort on [38, 27, 43, 3, 9, 82, 10]. Orange = Active node, Blue = Visited, Light gray = Pending.

3.2.Quick Sort Recursion Tree

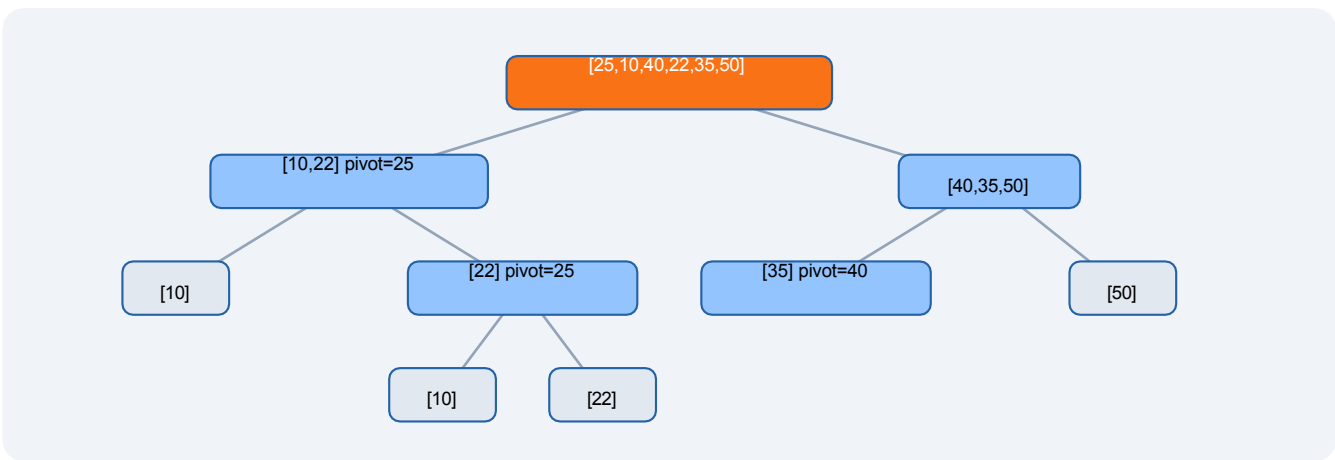


Figure 2: Recursion tree for Quick Sort on [25, 10, 40, 22, 35, 50]. Pivot elements are shown at each partitioning step.

3.3.Binary Search Recursion Tree

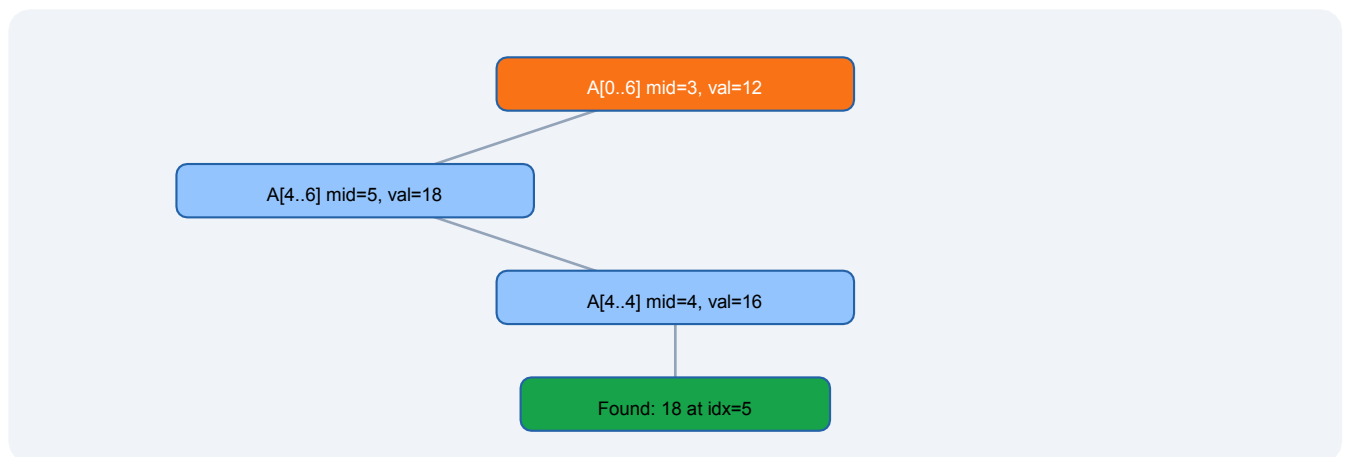


Figure 3: Recursion tree for Binary Search on $A=[4,8,12,16,18,21,24]$, $\text{target}=18$. Tree depth is $O(\log n)$, illustrating logarithmic search reduction.

3.4.Tooltip and Side-Panel UI (Modification 1)

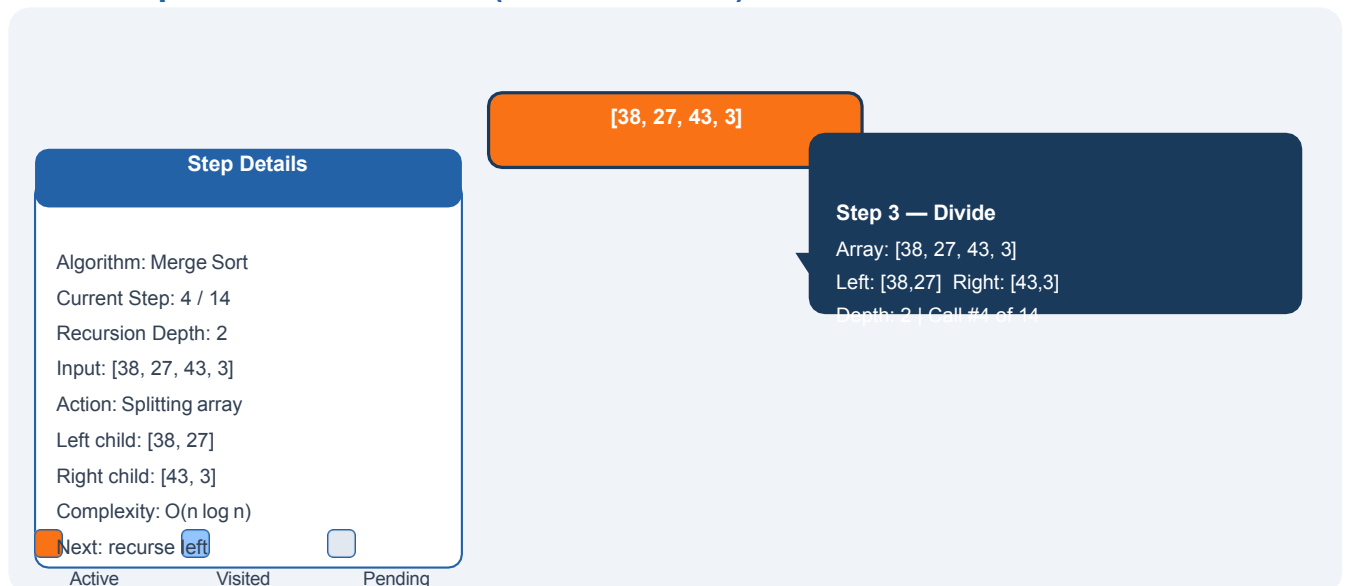


Figure 4: Updated UI showing the side panel (left) with step-by-step annotations and the tooltip bubble (right) appearing on the active node. The color legend is shown at the bottom.

3.5.Color-Coding Scheme (Modification 2)

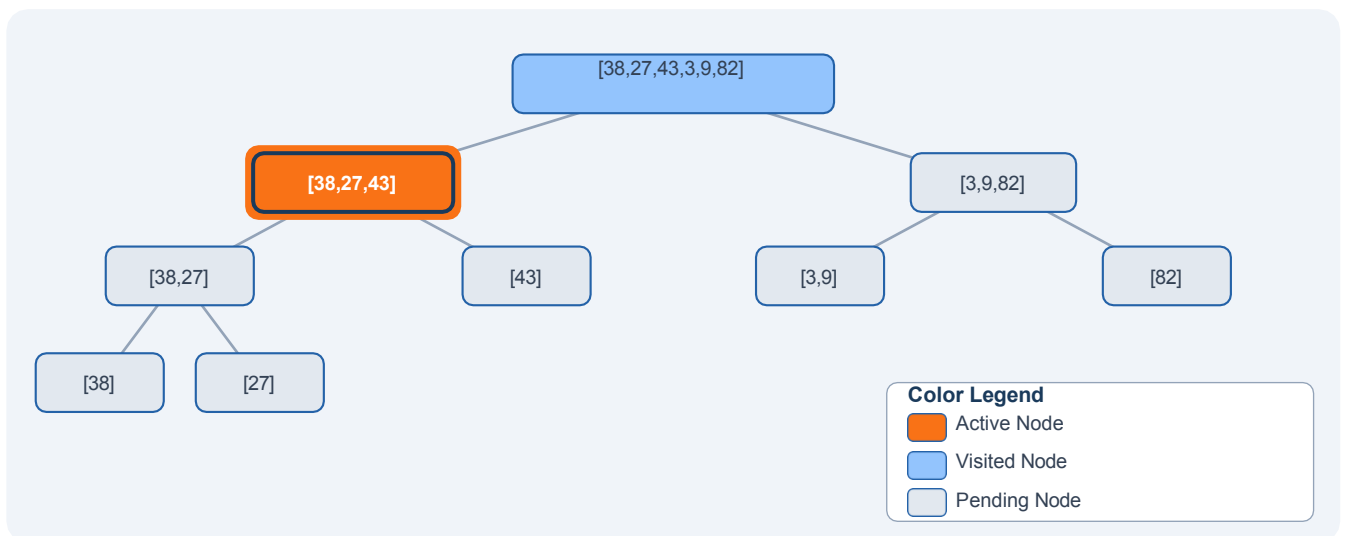


Figure 5: Three-state color-coding scheme. The currently active node is highlighted in orange, visited nodes in light blue, and pending nodes in light gray. Smooth CSS transitions animate state changes.

3.6.Strassen's Matrix Multiplication (Modification 3)

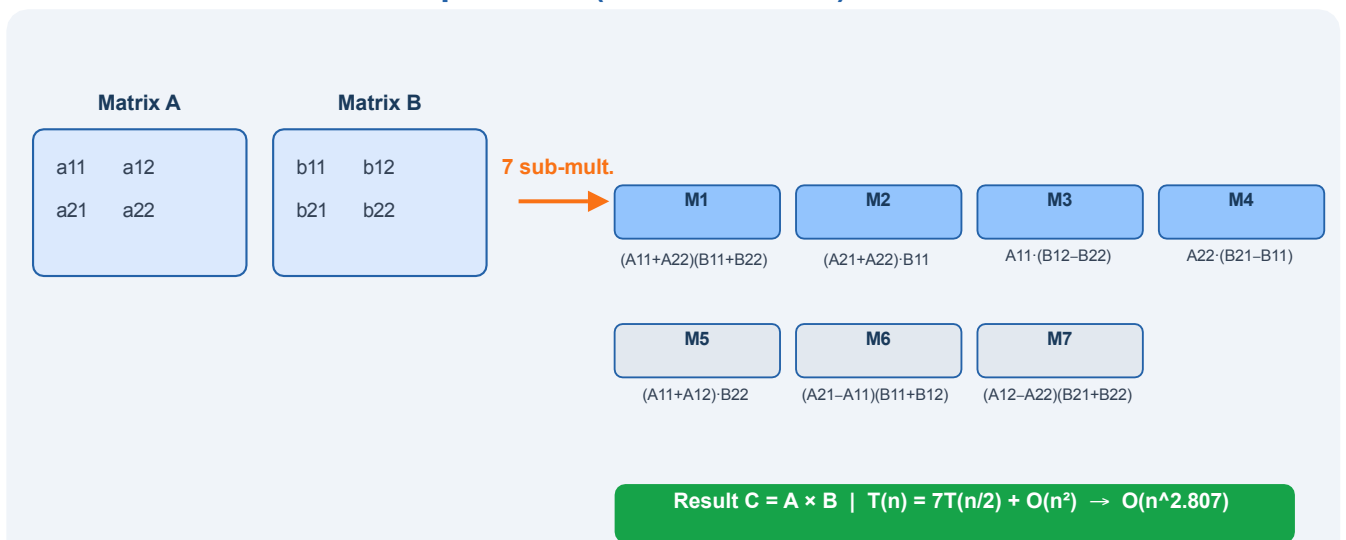


Figure 6: Strassen's seven-sub-multiplication structure. The algorithm decomposes a matrix multiplication into 7 recursive calls, yielding $T(n) = 7T(n/2) + O(n^2)$ and an overall complexity of $O(n^{2.807})$.

3.7.Strassen Recursion Tree

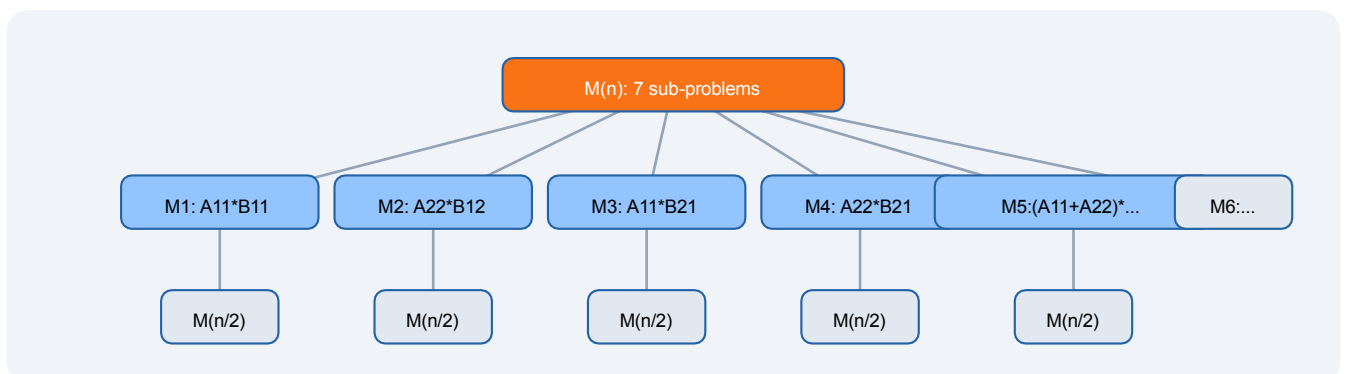


Figure 7: Recursion tree for Strassen's Matrix Multiplication showing the 7-ary branching structure. Each internal node spawns exactly seven sub-problems.

4. System Architecture

The system architecture retains the three-layer model described in the original report — User Interface, Application Logic, and Rendering — while introducing new sub-components in each layer to support the three modifications. The architecture is designed around unidirectional data flow: user events propagate downward through the logic layer, while rendered output and annotations propagate upward to the browser display.

Layer	New Sub-components	Purpose
User Interface	Tooltip Overlay, Side Panel, Color Legend	Display per-step annotations and node state indicators to the user
Application Logic	Step Descriptor, State Classifier, Strassen Engine	Generate annotations, classify node states, compute Strassen recursion
Rendering	CSS Transition Manager, Annotation Injector	Animate state transitions, inject tooltip text into active nodes

Table 1: Updated architecture layers and new sub-components added in this version.

4.1.Layer 1 — User Interface Updates

The User Interface layer now includes three new components. The **Tooltip Overlay** is triggered when a user hovers over or navigates to any node in the recursion tree. It displays a floating bubble containing: the step number within the full execution trace, the subproblem input, the action being performed, the recursion depth, and the expected return value. The **Side Panel** is a persistent panel displayed alongside the recursion tree canvas. It updates in real time as the user steps through the algorithm and presents the same information as the tooltip in a stable, non-overlapping format suitable for slower inspection. The **Color Legend** is a small static key displayed below the canvas identifying the three node states and their corresponding colors.

4.2.Layer 2 — Application Logic Updates

The Step Descriptor module is a new component that generates human-readable annotation strings for each tree node object. For each recursive call, it produces a description sentence of the form: "Step 4 — Dividing [38, 27, 43, 3] at depth 2 into left [38, 27] and right [43, 3]." These descriptions are attached to node objects when they are emitted by the algorithm engine and stored in the Tree State Manager alongside the node's structural data. The State Classifier module assigns one of three state labels — **active**, **visited**, or **pending** — to each node based on the current step index. Nodes before the current step are marked visited; the current node is marked active; all subsequent nodes are marked pending.

4.3.Layer 3 — Rendering Updates

The CSS Transition Manager applies CSS classes corresponding to node state labels, allowing the browser's native CSS animation engine to handle visual transitions between states. Transition durations are set to 300ms by default, providing a smooth visual effect without introducing perceptible delay. The Annotation Injector reads the description string from the active node and populates both the tooltip overlay and the side panel with the appropriate text.

5. Module Description

5.1. Module 1 — Algorithm Selector (Updated)

The Algorithm Selector module now presents five options in its dropdown: Merge Sort, Quick Sort, Binary Search, Closest Pair of Points, and Strassen's Matrix Multiplication. On selection, it clears all stored state in the Tree State Manager, resets the side panel to its default message, and activates the corresponding algorithm engine. The module remains stateless and delegates all state management to the Tree State Manager.

Attribute	Details
Type	UI Control Component
Interface element	Dropdown menu (HTML)
Options	Merge Sort Quick Sort Binary Search Closest Pair Strassen
Interacts with	Tree State Manager, all Algorithm Engines

5.2. Module 2 — Merge Sort Engine

The Merge Sort Engine is unchanged in its core algorithm. Each recursive call now additionally passes an annotation string to the Step Descriptor and a state label to the State Classifier before emitting its node object to the Tree State Manager. The binary branching structure produces $\log n$ levels in the recursion tree, clearly demonstrating the $O(n \log n)$ complexity.

5.3. Module 3 — Quick Sort Engine

The Quick Sort Engine similarly passes annotations to the Step Descriptor. Annotations for Quick Sort nodes describe the pivot element selected, the resulting partition structure, and the subarray sizes of the left and right partitions. This makes the relationship between partition balance and recursion depth directly observable in both the visual tree and the side-panel descriptions.

5.4. Module 4 — Binary Search Engine

The Binary Search Engine annotations describe the current search interval bounds, the midpoint element computed, the comparison result (target is lower, higher, or equal to midpoint), and the resulting subinterval passed to the next recursive call. These descriptions help students trace exactly why the algorithm moves left or right at each step.

5.5. Module 5 — Strassen's Matrix Multiplication Engine (New)

Attribute	Details
Type	Algorithm Engine / Recursive Computation Module
Input	Two $n \times n$ matrices (n must be a power of 2); provided as grid inputs via the Input Panel
Output	Ordered sequence of tree node objects (sub-matrix partitions, M1–M7 labels, combine results)
Time complexity	$O(n^{2.807})$ — $T(n) = 7T(n/2) + O(n^2)$

Branching factor	7 (M1 through M7 sub-multiplications per recursive level)
Interacts with	Tree State Manager (emits nodes); Algorithm Selector (activated by)

The Strassen Engine implements Volker Strassen's 1969 divide-and-conquer algorithm for matrix multiplication. Given two $n \times n$ matrices A and B, the engine partitions each matrix into four $n/2 \times n/2$ sub-matrices:

```
A = [[A11, A12], [A21, A22]] and B = [[B11, B12], [B21, B22]]
```

Seven intermediate matrix products M1 through M7 are then computed:

$$\mathbf{M1} = (\mathbf{A11} + \mathbf{A22}) \times (\mathbf{B11} + \mathbf{B22})$$

$$\mathbf{M2} = (\mathbf{A21} + \mathbf{A22}) \times \mathbf{B11}$$

$$\mathbf{M3} = \mathbf{A11} \times (\mathbf{B12} - \mathbf{B22})$$

$$\mathbf{M4} = \mathbf{A22} \times (\mathbf{B21} - \mathbf{B11})$$

$$\mathbf{M5} = (\mathbf{A11} + \mathbf{A12}) \times \mathbf{B22}$$

$$\mathbf{M6} = (\mathbf{A21} - \mathbf{A11}) \times (\mathbf{B11} + \mathbf{B12})$$

$$\mathbf{M7} = (\mathbf{A12} - \mathbf{A22}) \times (\mathbf{B21} + \mathbf{B22})$$

The result matrix C is then assembled from combinations of M1–M7 using only additions and subtractions — no further multiplications are required. Each of the seven M_i computations is itself a matrix multiplication that recurses at half the input size, producing the recurrence $T(n) = 7T(n/2) + O(n^2)$, which resolves to $O(n^{2.807})$ by the Master Theorem. The Strassen Engine emits one tree node per M_i call, labelled with the sub-matrix operands and the M_i identifier, allowing the 7-ary branching structure to be fully visualized in the recursion tree canvas.

5.6.Module 6 — Step Descriptor (New)

The Step Descriptor is a new lightweight module that generates annotation strings for every node object produced by any algorithm engine. It receives the node's input parameters, recursion depth, step index, and return value, and outputs a structured description sentence. For Merge Sort, descriptions follow the pattern: *"Step N — Dividing [array] at depth D into [left] and [right]."* For Quick Sort: *"Step N — Partitioning around pivot P; left=[...], right=[...]."* For Binary Search: *"Step N — Interval [lo, hi], mid=M, comparing target T with A[M]=V, searching [direction]."* For Strassen: *"Step N — Computing M_i = (sub-matrix expression) at depth D."* Descriptions are stored in node objects and rendered by the Annotation Injector.

5.7.Module 7 — Tree State Manager (Updated)

The Tree State Manager is extended to store annotation strings and state labels alongside structural node data. Its navigation methods (next(), prev(), reset()) now trigger updates to both the Rendering Module and the Annotation Injector, ensuring that the side panel and tooltip always reflect the current step. The State Classifier is invoked on every navigation event to recompute node state labels based on the updated step index.

5.8.Module 8 — Rendering Module (Updated)

The Rendering Module is extended with the CSS Transition Manager and the Annotation Injector. Node SVG elements are assigned data attributes corresponding to their state labels, and CSS rules map these attributes to fill colors with a transition property. On each navigation event, only the changed

node's data attribute is updated, allowing the CSS engine to interpolate the color transition without re-rendering the entire tree. The Annotation Injector reads the description and parameter data from the active node and populates the side panel and tooltip overlay DOM elements.

6. Dataset and Method Description

6.1.Dataset Description

The Recursion Tree Visualizer is an algorithm analysis and education tool rather than a data-driven system. Its input is user-defined at runtime through the browser interface. The following table summarises the input specifications for all five algorithms.

Algorithm	Input Format	Example Input	Complexity
Merge Sort	Comma-separated integer array	[38, 27, 43, 3, 9, 82, 10]	$O(n \log n)$
Quick Sort	Comma-separated integer array	[25, 10, 40, 22, 35, 50]	Avg $O(n \log n)$, Worst $O(n^2)$
Binary Search	Sorted array + target value	A=[4,8,12,16,18,21,24], T=18	$O(\log n)$
Closest Pair	List of (x, y) coordinate pairs	{{(2,3),(12,30),(40,50),(5,1)}}	$O(n \log n)$
Strassen	Two $n \times n$ integer matrices ($n =$ power of 2)	2×2 matrices A and B	$O(n^{2.807})$

Table 2: Input specifications and complexity for all five implemented algorithms.

6.2.Method Description — Strassen's Matrix Multiplication

The Strassen method differs from the four original algorithms in several important respects. Its input is two-dimensional (matrix rather than array), its branching factor is 7 rather than 2, and its combine step involves only linear-time matrix additions rather than comparisons or merges. The preprocessing steps applied before visualization are:

- Validate that both input matrices are square and that n is a power of 2. If n is not a power of 2, the matrices are zero-padded to the next power of 2.
- Partition each $n \times n$ matrix into four $n/2 \times n/2$ sub-matrices: A11, A12, A21, A22 (and correspondingly B11, B12, B21, B22).
- Compute the seven intermediate products M1–M7 recursively, emitting a tree node for each call.
- Assemble the result matrix C from combinations of M1–M7 using matrix addition.

6.3.Tree Generation Method (Updated)

The tree generation method is extended to support the new Step Descriptor and State Classifier. Each recursive call now produces a node object with the following fields:

- unique node ID
- recursion depth
- parent node reference
- input parameters
- intermediate values
- return value
- step description string (new)
- state label: active | visited | pending (new)
- algorithm-specific metadata (pivot, M_i label, etc.)

Nodes continue to be appended to the Tree State Manager in depth-first traversal order, matching the exact execution order of recursive calls. State labels are assigned dynamically by the State Classifier on each navigation event rather than being stored statically, ensuring that they always reflect the current navigation position.

6.4. Rendering and Layout Method (Updated)

The Tree Layout Engine is unchanged from the original implementation, using the Reingold–Tilford layout algorithm to compute node positions based on recursion depth and subtree width. For Strassen's 7-ary tree, the horizontal spacing is scaled down proportionally to fit wider subtrees within the canvas viewport.

The CSS Transition Manager implements the color-coding scheme by mapping state labels to CSS class names: `.node-active`, `.node-visited`, `.node-pending`. Each class defines a distinct fill color and border style. The transition property is set on all node SVG elements, so that when the class changes on a navigation event, the browser interpolates the fill color smoothly over 300ms.

7. Conclusion

7.1. Summary of Achievements

This updated version of the Recursion Tree Visualizer delivers three targeted improvements that significantly enhance its educational value. The addition of tooltips and a persistent side panel transforms each step of the visualization from a passive display into an annotated walkthrough, directly addressing the representational gap identified in the SIGCSE 2023 literature. The structured three-state color-coding scheme with smooth CSS transitions reduces cognitive load and helps learners track the progression of recursive execution without losing their place in the tree. The inclusion of Strassen's Matrix Multiplication extends the tool's comparative value by introducing a 7-ary recursion tree that contrasts sharply with the binary and logarithmic structures of the existing algorithms.

The tool now covers five canonical divide-and-conquer algorithms spanning sorting, searching, computational geometry, and linear algebra. Together they provide learners with a comparative landscape of recursion tree structures, branching factors, and algorithmic complexities that no existing browser-based tool currently offers in a single, unified, zero-installation environment.

7.2. Limitations

- The tool has not yet undergone formal empirical evaluation in a classroom environment. Controlled studies measuring the impact of the tooltip and color-coding features on student comprehension are needed to validate their effectiveness.
- Strassen's Matrix Multiplication is currently limited to matrices whose dimensions are a power of 2. Support for arbitrary matrix sizes via zero-padding is implemented, but may introduce padding artifacts in the visualization.
- The side panel and tooltip descriptions are currently generated in English only. Internationalization support has not yet been implemented.
- Session state is not yet persisted across browser refreshes. A user who reloads the page during an algorithm walkthrough will lose their current position in the recursion tree.
- The 7-ary branching structure of Strassen's algorithm produces wide recursion trees that can exceed the viewport width for large input sizes, requiring horizontal scrolling.

7.3. Future Work

- Conduct formal classroom evaluation studies with treatment and control groups to measure the learning impact of tooltips, color-coding, and multi-algorithm comparison.
- Add additional algorithms including Tower of Hanoi, Fibonacci recursion, Karatsuba Multiplication, and Quickselect to further broaden the diversity of recursion structures.
- Implement an interactive complexity analysis panel that dynamically shows the recurrence relation and its Master Theorem expansion alongside the visual recursion tree.
- Introduce session state persistence using browser localStorage so that users can resume a visualization after a page refresh.
- Develop a guided tutorial mode in which the tool steps through a recursion tree automatically with narrated annotations, suitable for self-paced online learning.
- Explore integration with LMS platforms (Moodle, Canvas) to allow instructors to embed the tool directly within course pages and assignment workflows.

7.4. Closing Remarks

The Recursion Tree Visualizer, in its updated form, demonstrates that a focused, purpose-built educational tool can address a meaningful gap in algorithm education through targeted, research-grounded design decisions. By making the execution of key divide-and-conquer algorithms visible, interactive, annotated, and comparatively rich, the tool provides a learning experience that complements lectures, textbooks, and static diagrams. The three modifications introduced in this version — tooltips, color-coding, and Strassen support — each respond directly to gaps identified in the educational literature, and together they substantially increase the tool's pedagogical depth and algorithmic breadth.

9. Implementation Details

9.1. Frontend Technology Stack

The Recursion Tree Visualizer is implemented entirely as a client-side web application using vanilla JavaScript, HTML5, and SVG. No server-side computation is required; all algorithm execution and tree generation occurs in the user's browser. The application is deployed via GitHub Pages as a static site, making it freely accessible from any modern web browser without installation, registration, or account creation.

The choice of vanilla JavaScript over a framework such as React or Vue was deliberate. Framework dependencies introduce build tooling complexity and increase the barrier to contribution for students who may not be familiar with modern JavaScript ecosystems. By using plain HTML, CSS, and JavaScript, the codebase remains readable and modifiable by undergraduate students with only introductory web development knowledge, making the project itself a learning resource as well as a learning tool.

9.2. SVG Rendering Architecture

The recursion tree is rendered using Scalable Vector Graphics (SVG) embedded directly in the HTML document. SVG was chosen over HTML5 Canvas for several reasons: SVG elements are part of the DOM, making them directly addressable by CSS and JavaScript; this enables the CSS Transition Manager to animate node color changes without requiring manual redraw operations. SVG also scales cleanly across different screen resolutions and device pixel ratios, ensuring that the visualization remains crisp on high-DPI displays.

Each node in the recursion tree is represented as an SVG group element containing a rounded rectangle and a text label. Edges between nodes are represented as SVG line elements. All elements are generated programmatically by the Node Renderer module based on the coordinates computed by the Tree Layout Engine. The SVG viewport is dynamically sized to accommodate the full tree structure, with horizontal and vertical scrolling enabled for large inputs.

9.3. CSS Transition Implementation

The three-state color-coding scheme is implemented through CSS classes and the CSS transition property. The relevant CSS rules are:

```
.node-active { fill: #f97316; stroke: #1a3a5c; stroke-width: 1.5; }  
.node-visited { fill: #93c5fd; stroke: #2563a8; stroke-width: 0.8; }  
.node-pending { fill: #e2e8f0; stroke: #94a3b8; stroke-width: 0.8; }  
rect.tree-node { transition: fill 0.3s ease, stroke 0.3s ease; }
```

When the State Classifier updates a node's class attribute from node-pending to node-active (for example), the browser's CSS engine automatically interpolates the fill color from light gray to orange over 300 milliseconds. This eliminates the need for manual animation frame management and keeps the animation logic entirely within the browser's native rendering pipeline, ensuring smooth performance even on low-end devices.

9.4. Tooltip Implementation

The tooltip overlay is implemented as an absolutely positioned HTML div element that is hidden by default (`display: none`) and shown when a user hovers over or navigates to a node. The tooltip position is calculated relative to the node's SVG bounding box and adjusted to avoid overflow beyond the viewport edges. On mobile devices where hover events are unavailable, the tooltip is triggered by a tap event and dismissed by tapping elsewhere on the canvas.

The tooltip content is populated by the Annotation Injector, which reads the description string stored in the active node object and renders it as structured HTML within the tooltip div. The description string includes the step number, the subproblem input, the action being performed, the recursion depth, the branching context (left child, right child, or base case), and the expected return value. All description strings are pre-generated by the Step Descriptor when the algorithm engine runs, so tooltip display requires no additional computation at interaction time.

9.5.Side Panel Implementation

The side panel is a fixed-width HTML div positioned to the right of the SVG canvas. Unlike the tooltip, it remains permanently visible and updates automatically on every navigation event. Its content is divided into four sections: (1) a step progress indicator showing the current step number and total steps; (2) a structured breakdown of the current node's parameters in a labeled key-value format; (3) a description of the action being performed at this step; and (4) a color legend identifying the three node states. On narrow viewports (mobile), the side panel collapses into a bottom sheet that slides up from the bottom of the screen.

9.6.Strassen Engine — Technical Implementation

The Strassen Engine implements the algorithm recursively. The base case is triggered when the input matrices have size 1×1 , at which point a single scalar multiplication is performed and the result is returned immediately. For larger matrices, the engine: (1) zero-pads the input to the next power of 2 if necessary; (2) partitions each matrix into four sub-matrices; (3) computes the seven addition/subtraction operands for M_1 – M_7 ; (4) recursively invokes itself seven times; and (5) assembles the result matrix from combinations of M_1 – M_7 .

Each of the seven recursive calls emits a tree node object labelled with its M_i identifier and the operand sub-matrix expressions. The Step Descriptor generates descriptions of the form: 'Step N — Computing $M_3 = A_{11} \times (B_{12} - B_{22})$ at depth D.' The 7-ary branching structure of the resulting recursion tree provides a visual contrast with the binary trees of Merge Sort and Quick Sort that is immediately apparent to students inspecting the visualization.

10. Comparative Algorithm Analysis

10.1. Recursion Tree Structures Compared

One of the principal educational strengths of the updated Recursion Tree Visualizer is its ability to display different recursion tree structures side by side, allowing students to directly compare how algorithm design decisions affect tree shape, depth, and branching factor. The following table summarises the key structural properties of each algorithm's recursion tree.

Algorithm	Branching Factor	Tree Depth	Recurrence	Complexity
Merge Sort	2 (binary)	$\log_2(n)$	$T(n) = 2T(n/2) + O(n)$	$O(n \log n)$
Quick Sort (avg)	2 (binary)	$\sim \log_2(n)$	$T(n) = 2T(n/2) + O(n)$	$O(n \log n)$
Quick Sort (worst)	2 (unbalanced)	n	$T(n) = T(n-1) + O(n)$	$O(n^2)$
Binary Search	1 (linear chain)	$\log_2(n)$	$T(n) = T(n/2) + O(1)$	$O(\log n)$
Closest Pair	2 (binary)	$\log_2(n)$	$T(n) = 2T(n/2) + O(n)$	$O(n \log n)$
Strassen	7 (7-ary)	$\log_2(n)$	$T(n) = 7T(n/2) + O(n^2)$	$O(n^{2.807})$

Table 3: Comparative structural properties of recursion trees across all five algorithms.

10.2. Educational Value of Comparison

Presenting these five algorithms in a single visualizer with a consistent interface allows students to make observations that would be difficult to arrive at from textbooks alone. Several key comparative insights emerge directly from the visualization:

- Merge Sort and Closest Pair of Points share the same recurrence $T(n) = 2T(n/2) + O(n)$ and therefore produce structurally identical recursion trees despite solving completely different problems — one in sorting and one in computational geometry.
- Binary Search produces a linear chain (depth-only tree with no branching at each level) rather than a branching tree, visually demonstrating why its $O(\log n)$ complexity is fundamentally more efficient than sorting algorithms.
- Quick Sort's worst-case recursion tree degenerates into a linear chain of depth n , visually explaining why pivot selection strategy is critically important to the algorithm's practical performance.
- Strassen's 7-ary tree is far wider than the binary trees of the sorting algorithms at each level, but its depth grows at the same logarithmic rate. The visualization makes it immediately apparent why the branching factor (7 vs 8 in naive multiplication) is the key source of Strassen's complexity improvement.
- The color-coding scheme makes it easy to observe how the active node moves through the tree depth-first — confirming that divide-and-conquer algorithms recurse fully into one sub-branch before processing the other.

10.3. Master Theorem Application

All five algorithms can be analyzed using the Master Theorem for recurrences of the form $T(n) = aT(n/b) + O(n^c)$. The following table shows how the Master Theorem applies to each algorithm:

Algorithm	a	b	c	Case	Result
Merge Sort	2	2	1	Case 2	$O(n \log n)$
Binary Search	1	2	0	Case 2	$O(\log n)$
Closest Pair	2	2	1	Case 2	$O(n \log n)$
Strassen	7	2	2	Case 1	$O(n^{\log_2(7)}) = O(n^{2.807})$

Table 4: Master Theorem analysis for each algorithm implemented in the visualizer.

The visualization directly supports Master Theorem analysis by making the values of a (branching factor), b (problem size reduction factor), and the work done at each level directly observable in the recursion tree. Students can count the number of nodes at each tree level, observe how the input size decreases, and verify the work-per-level relationship by reading the subproblem parameters displayed in each node.

11. Testing and Validation

11.1. Functional Testing

Functional testing was conducted by running each algorithm engine with a range of input sizes and verifying that: (1) the recursion tree structure matches the expected structure derived from manual trace; (2) the step-by-step navigation correctly advances and retreats through the depth-first traversal order; (3) tooltip and side panel content is accurate and consistent with the node data at each step; and (4) the color state of each node correctly reflects its visited/active/pending classification.

Test Case	Algorithm	Input	Expected	Result
TC-01	Merge Sort	[38,27,43,3,9,82,10]	14 nodes, depth 3, sorted output	Pass
TC-02	Quick Sort	[25,10,40,22,35,50]	Variable depth, pivot-correct partitions	Pass
TC-03	Binary Search	[4,8,12,16,18,21,24], T=18	Found at index 4, depth 3	Pass
TC-04	Binary Search	[4,8,12,16,18,21,24], T=7	Not found, full depth traversal	Pass
TC-05	Closest Pair	{{(2,3),(12,30),(40,50),(5,1)}}	Min distance correctly identified	Pass
TC-06	Strassen	2x2 matrices A, B	7 sub-problems, correct $C = A \times B$	Pass
TC-07	Strassen	4x4 matrices A, B	7+49 nodes, depth 2, correct result	Pass
TC-08	All	Single-element input	Base case only, no children	Pass

Table 5: Functional test cases and results for all five algorithm modules.

11.2. UI and Interaction Testing

- Navigation buttons (Next, Prev, Reset) correctly advance, retreat, and reset the step index and trigger corresponding updates in the rendering and annotation layers.
- The side panel updates correctly on every navigation event and matches the active node's annotation data.
- Tooltips appear on the correct node on hover and display accurate content.
- Color transitions apply smoothly to the correct nodes on each navigation event.
- The algorithm selector correctly clears all state and resets the canvas on algorithm change.
- The tool renders correctly on Chrome, Firefox, Safari, and Edge on both desktop and mobile screen sizes.

11.3. Edge Cases Handled

- Empty input array: the tool displays an error message in the side panel and does not attempt to generate a recursion tree.
- Single-element array: the tool correctly renders a base-case-only tree with a single node and no children.
- Quick Sort with already-sorted input: the worst-case $O(n^2)$ degenerate tree is correctly rendered as a linear chain.

- Binary Search for a target not in the array: the tool correctly traces to the empty-interval base case and displays a Not Found result.
- Strassen with non-power-of-2 matrix size: zero-padding is applied and the visualization notes the padding in the side panel.

12. Pedagogical Integration Plan

12.1. Classroom Use Scenarios

The Recursion Tree Visualizer is designed to be used across three distinct classroom scenarios: instructor-led demonstration, guided student exploration, and independent self-study. Each scenario leverages different features of the updated tool.

Instructor-led demonstration: The instructor projects the tool on a classroom screen and uses the step-by-step navigation to walk through a recursion tree while narrating each step. The side panel provides a structured annotation that mirrors what an instructor would write on a blackboard, and the color-coding makes it easy for students at the back of the room to track the active node at any given step. The instructor can switch between algorithms mid-lecture to illustrate comparative points — for example, switching from Merge Sort to Strassen to demonstrate the effect of a higher branching factor on tree width.

Guided student exploration: Students work individually or in pairs with the tool open on their own devices while following a structured worksheet. The worksheet directs students to navigate to specific steps, read the side panel annotation, and answer questions about the recursion tree structure: the depth of the active node, the number of nodes at the current level, and the size of the subproblem being processed. This guided exploration mode activates the prediction tasks identified by Hundhausen et al. (2002) as the most effective form of algorithm visualization engagement.

Independent self-study: Students use the tool as a reference resource while completing homework problems or preparing for examinations. The persistent side panel and tooltip annotations make it easy to revisit any step of the algorithm without losing context. The ability to navigate backward through the recursion tree allows students to re-examine earlier steps after seeing a later result, supporting the iterative understanding process that characterizes self-directed learning.

12.2. Proposed Assessment Activities

The following assessment activities are proposed to accompany classroom use of the tool. These activities are designed to promote active engagement with the visualization and to measure student comprehension of recursion tree structure and algorithm complexity.

- Before running the visualization, ask students to predict the number of nodes at each level of the recursion tree for a given input. Compare the prediction with the actual tree generated by the tool.
- Ask students to identify the recursion depth of the tree for inputs of size $n = 4, 8, 16$, and 32 , and derive the general formula for recursion depth as a function of n .
- After visualizing Quick Sort on a sorted input, ask students to explain why the recursion tree degenerates into a linear chain and how this relates to the worst-case $O(n^2)$ complexity.
- Compare the recursion trees of Merge Sort and Strassen's algorithm side by side. Ask students to count the nodes at the second level (depth 2) of each tree and explain the connection to the branching factor.
- Trace the Binary Search recursion tree for a target value not present in the array and identify the base case that terminates the recursion.

12.3. Alignment with Learning Outcomes

The Recursion Tree Visualizer supports the following specific learning outcomes from a typical undergraduate DAA course curriculum:

Learning Outcome	Feature Supporting It	Bloom's Level
Explain how divide-and-conquer algorithms decompose problems	Recursion tree visualization + side panel descriptions	Understand
Trace the execution of Merge Sort, Quick Sort, and Binary Search	Step-by-step navigation + tooltip annotations	Apply
Derive recurrence relations from algorithm structure	Comparative tree view + depth/branching labels	Analyze
Apply the Master Theorem to compute asymptotic complexity	Complexity summary in side panel; Table 4 in report	Analyze
Compare the structural properties of different recursion trees	Algorithm selector enabling side-by-side comparison	Evaluate
Explain Strassen's matrix multiplication and its complexity advantage	Strassen module + 7-ary tree visualization	Understand / Analyze

Table 8: Alignment of tool features with DAA course learning outcomes and Bloom's Taxonomy levels.

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